EQUIPHASE DEQ: STRUCTURAL PRESERVATION AND MULTISTABLE ENERGY LANDSCAPES IN REAL MOLECULAR SYSTEMS

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ABSTRACT

Deep equilibrium models (DEQs) typically purchase convergence guarantees through monotone-operator or contraction constraints that force a unique fixed point. We study the autonomous use of equilibrium layers — relaxation dynamics whose attractors are themselves the object of interest, as in energy landscapes and associative memories — and argue that in this setting the uniqueness certificate is not free: physical systems are multistable, and a map that provably admits one attractor cannot represent two metastable states, let alone their relative free energies. We introduce EquiPhase DEQ, an equilibrium model whose dynamics are damped velocity Verlet steps on a learned conservative potential with periodicity built into the architecture; the step is conformally symplectic ($J^\top \Omega J = (1-\eta)\Omega$), so it dissipates volume at a known rate while inheriting the geometric stability of symplectic integration. On a 32-dimensional synthetic double-well system with known analytical structure, the model passes a preregistered battery of structural and empirical gates. We then evaluate on real molecular data: 750,000 frames of alanine dipeptide backbone dihedrals. Under a preregistered protocol with sealed evaluation scripts, EquiPhase recovers the β and α_R metastable basins in 5/5 independent seeds, with attractor locations within $\pm 25^\circ$ of data-derived anchors and the correct free-energy ordering, and additionally locates the rare α_L state that occupies only 3.0% of the data in 5/5 seeds. We report our negative results in full: an unconstrained score-matching baseline also locates the basins, so the contribution of structure is not localization but the existence of a consistent scalar potential (the baseline field has non-zero curl and admits no free-energy readout), while a verified Banach contraction baseline collapses to a single non-physical fixed point exactly as theory predicts. All quantitative claims trace to hash-sealed, preregistered evaluation runs.

1 INTRODUCTION

Deep equilibrium models express implicit layers through fixed-point solutions (Bai et al., 2019), within the broader family of implicit and differentiable-optimization layers (Chen et al., 2018; Amos & Kolter, 2017; Agrawal et al., 2019; El Ghaoui et al., 2021). To guarantee that the fixed point exists, is unique, and is reachable, standard formulations impose global contraction or monotone-operator structure (Winston & Kolter, 2020; Revay et al., 2020). In physical dynamics — protein conformational transitions, hysteretic switches, multistable landscapes generally — this uniqueness certificate is not a convenience but a contradiction: by Banach's fixed-point theorem (Banach, 1922), a globally contractive layer has exactly one attractor and therefore cannot represent two coexisting metastable states, let alone the barrier between them. We call this failure mode *basin collapse*. It is not an implementation bug; it is a direct consequence of demanding global contraction, and Section 6 exhibits it on real molecular data with a contraction that is verified, not assumed.

Single-mode collapse is a live problem elsewhere. Recent experiment-guided ensemble modelling is motivated by the observation that the flagship sequence-conditioned structure predictor collapses

to a single dominant conformation even when the underlying structure is heterogeneous, and repairs this by injecting experimental likelihoods during sampling (Maddipatla et al., 2026). That collapse arises from a training objective and in an input-conditioned setting, not from a fixed-point certificate in the autonomous regime; we cite it not as an instance of Corollary 1 but as evidence that single-mode collapse under genuine multistability is costly in practice, and that the remedies currently on offer are data-side rather than architecture-side.

Scope: autonomous equilibrium dynamics, not input-conditioned inference. We state the boundary of our claim precisely, because the standard DEQ and the model studied here use fixed points for different purposes. A conventional DEQ solves an input-conditioned problem $z^* = f_\theta(z^*, x)$: contraction guarantees uniqueness per input x, and multimodality across inputs remains representable through the dependence of z^* on x. Our subject is the autonomous regime in which the equilibrium map is iterated as relaxation dynamics on a state space and the attractors themselves carry the semantics — the regime of energy-based attractor networks (Ramsauer et al., 2021; Krotov & Hopfield, 2016) and of physical metastability, where "which basin does this initial condition fall into, and how deep is it?" is the question being asked. In this regime the per-input uniqueness certificate becomes a global one, and Corollary 1 applies with full force (Remark 1 makes the distinction formal). All claims in this paper are claims about the autonomous regime.

Contributions.

- **A multistable equilibrium architecture.** EquiPhase DEQ couples a learned conservative potential (periodicity guaranteed by construction) with damped velocity Verlet dynamics, deliberately forgoing global fixed-point uniqueness to retain multistability. The novelty is not conservative score parameterization per se — gradient-of-energy score models are established (Salimans & Ho, 2021; Saremi et al., 2018) — but the use of a conformally symplectic integrator (Proposition 1) as the equilibrium solver, which makes fixed points of the model coincide exactly with critical points of the learned potential and attractor counting coincide with metastable-state counting.
- **An impossibility observation made empirical.** We verify at run time that our monotone baseline is a genuine contraction (theoretical bound 0.95; empirical Lipschitz maximum $0.788 < 1$) and observe the Banach-mandated collapse: all 576 grid initializations converge to a single fixed point at $(\phi, \psi) = (-50.1°, +18.5°)$ — a location corresponding to no metastable state of the data — with training loss frozen at 0.712 versus 0.019 for EquiPhase.
- **Real-data validation with preregistered gates.** On alanine dipeptide (750k frames), EquiPhase passes location, identity, and free-energy-ordering gates in 5/5 seeds and detects the rare α_L state (3.0% occupancy) in 5/5 seeds — a state our own coarse density EDA failed to isolate.
- **Honest baselines and full disclosure.** A vanilla score network matches basin locations (our preregistered gate R4b therefore fails, and we report it); its field, however, has mean absolute curl 1.67 and thus no consistent potential. The differentiating claim is calibrated: structure buys a free-energy readout, not localization. All experiments follow a preregistration with sealed scripts and hash-anchored data; one auditor-side implementation error (an invalid contraction in the first monotone run) is documented and corrected in the record.

2 Related Work

Deep Equilibrium Models and Monotone Constraints. DEQs (Bai et al., 2019; 2020) reformulate deep networks by solving for implicit fixed points $z^* = f_\theta(z^*, x)$ rather than evaluating explicit feedforward layers; neural ODEs are the continuous-depth counterpart (Chen et al., 2018; Dupont et al., 2019). Forward passes rely on classical accelerated fixed-point solvers (Anderson, 1965; Broyden, 1965), and training uses implicit differentiation or its inexact and Jacobian-free variants (Geng et al., 2021; Fung et al., 2022); the framework has been deployed, e.g., in inverse problems in imaging (Gilton et al., 2021). To guarantee global fixed-point existence and numerical convergence during forward and backward passes, subsequent architectures enforce strict structural constraints rooted in monotone operator theory (Ryu & Boyd, 2016). Monotone operator equilibrium networks (Winston & Kolter, 2020) enforce strong monotonicity, and Lipschitz-certified

equilibrium models (Revay et al., 2020) parameterize networks with prescribed Lipschitz bounds; Jacobian regularization stabilizes the solver without a hard certificate (Bai et al., 2021). In the contractive regime ($\mathrm{Lip} < 1$) these guarantees are obtained, but Banach's theorem then restricts the autonomous iteration of the map to a single global attractor ($N = 1$). Consequently, contractive DEQs used as relaxation dynamics are structurally incapable of representing multistable physical systems or mapping energy barriers between distinct metastable states; the per-input conditional setting is unaffected (Remark 1).

Conservative Score Parameterization and Energy-Based Models. Parameterizing a score as the gradient of a scalar energy predates this work: energy-based models define densities through scalar energies (LeCun et al., 2006; Du & Mordatch, 2019; Grathwohl et al., 2020; Song & Kingma, 2021), score matching itself was introduced for unnormalized energy models (Hyvärinen, 2005; Song et al., 2019; Vincent, 2011), deep energy estimator networks learn V_θ with $S = -\nabla V_\theta$ (Saremi et al., 2018), denoising diffusion models train unconstrained score/noise predictors at scale (Ho et al., 2020), and Salimans & Ho (2021) directly ask whether EBMs should model the energy or the score, finding the constrained parameterization competitive. We therefore do not claim conservativeness as a novel mechanism. Our contribution relative to this line is what the conservative field is *for*: it is embedded as the force term of a dissipative equilibrium layer whose fixed points are the critical points of V_θ, turning the energy into an executable attractor model with a free-energy readout, and this combination is evaluated under an adversarial preregistered protocol against both an unconstrained score field (which localizes but cannot integrate) and a certified contraction (which cannot localize at all).

Physics-Informed Equilibrium and Conservative Networks. To incorporate physical structure into neural representations, Hamiltonian Neural Networks (Greydanus et al., 2019), Lagrangian Neural Networks (Cranmer et al., 2020), Symplectic ODE-Net (Zhong et al., 2020), and symplectic recurrent networks (Chen et al., 2020) learn scalar functions whose derivatives govern conservative time evolution; physics-informed learning more broadly constrains networks with governing equations (Raissi et al., 2019; Karniadakis et al., 2021). Classical HNNs operate primarily as explicit ODE integrators rather than implicit equilibrium models, and require full phase-space position and momentum pairs during training. Our integrator choice draws on the geometric numerical integration literature: velocity Verlet is the workhorse of molecular dynamics (Verlet, 1967; Leimkuhler & Matthews, 2015), and its structural stability under discretization is classical (Hairer et al., 2006); conformally symplectic (fixed-rate dissipative) dynamics have likewise been exploited for optimization (França et al., 2020). EquiPhase DEQ extends conservative potential learning into the implicit equilibrium setting by coupling periodic neural potentials with damped velocity Verlet steps, replacing exact conservation with a controlled dissipative structure ($J^\top \Omega J = (1-\eta)\Omega$) so that trajectories converge to attractors while inheriting the geometric stability of symplectic integration.

Score-Based Modeling and Molecular Energy Landscapes. Denoising Score Matching (Song & Ermon, 2019; Song et al., 2021) estimates the score field $\nabla_q \log p(q)$ without computing intractable partition functions. Machine learning now permeates molecular simulation (Noé et al., 2020; Hollingsworth & Dror, 2018): neural interatomic potentials (Behler & Parrinello, 2007; Schütt et al., 2017) and coarse-grained force fields that learn free-energy surfaces (Wang et al., 2019) target the energetics, while structure predictors resolve the fold itself (Jumper et al., 2021). On the kinetics side, slow collective variables are identified by TICA (Pérez-Hernández et al., 2013), Boltzmann generators sample equilibrium ensembles with exact reweighting (Noé et al., 2019), and Markov state modeling (Chodera & Noé, 2014; Scherer et al., 2015) and its deep variants such as VAMPnets (Mardt et al., 2018; Wu & Noé, 2020) identify metastable states and their interconversion kinetics. A parallel line makes ensembles *experiment-consistent* by injecting differentiable likelihoods for NMR, electron-density and cryo-EM observables as guidance gradients into a pretrained structure predictor (Maddipatla et al., 2026). That work is explicit that its ensembles are explanatory with respect to the supplied measurements and the model prior rather than calibrated thermodynamic populations, that mixture weights should not be read as equilibrium populations, and that energy-based or maximum-entropy population estimation remains the open direction. EquiPhase addresses the complementary half of that gap: not which structures are consistent with a given measurement, but whether the equilibrium layer itself can carry a consistent scalar landscape at all, evaluated on a system where the free-energy ordering can be scored rather than assumed. While unconstrained

score networks can accurately fit local density gradients, their estimated vector fields generally exhibit non-zero curl ($\nabla \times S \neq 0$), rendering them non-conservative: no single scalar potential $V(q)$ has $-\nabla V = S$ exactly, and any potential extracted post hoc (e.g., by Helmholtz projection or path integration) depends on the projection or path chosen. EquiPhase DEQ removes this ambiguity by constructing $S(q) = -\nabla_q V_\theta(q)$ structurally, guaranteeing zero curl and a valid free-energy readout. Section 7 positions the present contribution relative to the Markov-state-modeling toolchain explicitly.

Attractor Networks. Attractor dynamics as computation originates with Hopfield (1982); modern Hopfield networks (Ramsauer et al., 2021; Krotov & Hopfield, 2016) realize exponential-capacity associative memories as attractor dynamics and are the closest machine-learning precedent for our autonomous regime; they target pattern retrieval rather than physical free-energy landscapes, and their update rules are not integrators of a learned Hamiltonian-type potential.

3 METHOD: EQUIPHASE DEQ

Learned potential with structural guarantees. EquiPhase DEQ represents a system by a scalar potential and reaches equilibria by integrating damped conservative dynamics. We describe the model in three deliberately separated layers, so that the reader can always tell which properties are *guaranteed by construction*, which are *inherited from an analytical prior*, and which are *empirical*. (i) *Structure:* the potential is a function of an embedding chosen so that required invariances hold exactly — in the synthetic system, the symmetry and identity properties are architectural (gates G1/G2′); on dihedral data, the embedding $[\sin\phi, \cos\phi, \sin\psi, \cos\psi]$ makes V_θ exactly 2π-periodic. (ii) *Prior:* where an analytical potential V_{base} exists (synthetic system) we learn a residual, $V_{\text{total}} = V_{\text{base}} + v_{\text{net}}$; where none exists (molecular data) V_θ is learned alone, and we disclose that the prior layer is only partially applicable. (iii) *Empirics:* all remaining claims are established by preregistered gates.

Damped velocity Verlet dynamics. States are phase-space pairs (q, p). One step of the model applies

$$q_{1/2} = q_t + \tfrac{\Delta t}{2m}\, p_t, \tag{1}$$

$$p_{t+1} = (1-\eta)\, p_t - \Delta t\, \nabla_q V(q_{1/2}), \qquad \eta = \gamma \Delta t/m, \tag{2}$$

$$q_{t+1} = q_{1/2} + \tfrac{\Delta t}{2m}\, p_{t+1}. \tag{3}$$

With $\eta = 0$ this is the classical velocity Verlet integrator; with $\eta \in (0, 1)$ it contracts phase-space volume at a fixed, known rate while retaining the integrator's geometric stability. Equilibria of the map are exactly the pairs $(q^\star, 0)$ with $\nabla_q V(q^\star) = 0$: fixed points of the dynamics coincide with critical points of the learned potential, so counting attractors of the model is counting metastable states of V.

What the learned V estimates. Because V_θ is trained by denoising score matching, the population optimum of $-\nabla V_\theta$ is the score of the σ-smoothed data density; the learned V is therefore an estimate of the (smoothed) free-energy surface over the chosen collective variables, $V \approx -k_B T \log p_\sigma$, not of a microscopic potential energy. Gate R3 is accordingly a claim about the free-energy landscape in (ϕ, ψ) — the object of interest in metastability analysis — and not about physical information beyond the data distribution (disclosure D8′). What the surface adds over a binned density (executability, curl-freeness, and resolution where mass is scarce) is taken up in Section 7.

4 WHY UNIQUENESS GUARANTEES EXCLUDE MULTISTABILITY

Proposition 1 (Dissipative structure)**.** *Let J be the Jacobian of one damped velocity Verlet step* (1)–(3) *with damping η, and Ω the canonical symplectic form. Then $J^\top \Omega J = (1-\eta)\,\Omega$.*

Proof. Appendix A; the identity is a direct computation with the symmetric Hessian $H = \nabla^2 V(q_{1/2})$, and holds for every state, i.e., globally and independently of the learned V. □

The step is therefore not symplectic (it does not conserve energy — by design) but *conformally* so (França et al., 2020): phase-space volume shrinks by the constant factor $(1 - \eta)$ per step, which is what drives trajectories into the basins of V rather than orbiting them. In the synthetic system this identity is additionally verified numerically to the reported gate tolerance rather than assumed.

Proposition 2 (Banach collapse)**.** *If a DEQ update map g is a contraction,* $\mathrm{Lip}(g) \leq c < 1$*, on a complete metric space, then g admits exactly one fixed point, and every initialization converges to it.*

Corollary 1. *A DEQ whose well-posedness is certified via global contraction (or strong monotonicity implying it), when iterated as autonomous relaxation dynamics, has attractor count $N = 1$ independent of the data it was trained on. Representing $N \geq 2$ metastable states — avoiding basin collapse — requires forgoing the global uniqueness certificate.*

Remark 1 (Input-conditioned DEQs are not the target of this claim)**.** *For a conditional map $g_x(z) = f_\theta(z, x)$ with $\mathrm{Lip}_z(g_x) \leq c < 1$, Banach's theorem yields one fixed point per input x; the solution map $x \mapsto z^*(x)$ may still be richly multimodal across inputs. Corollary 1 concerns the autonomous regime of Section 1, where a single map is iterated on the state space and the attractor set itself is the model's output. The molecular task is intrinsically of this kind: metastable states are properties of one landscape, not of an input–output map.*

Remark 2 (The contraction baseline is a diagnostic, not a straw man)**.** *Contraction and strong monotonicity are not exotic choices; they are the default well-posedness prescription of the DEQ literature (Winston & Kolter, 2020; Revay et al., 2020), the certificates a careful practitioner is instructed to install before trusting an equilibrium layer at all. Our claim is that this standard prescription, transplanted unmodified into the autonomous regime, fails* silently*: the collapsed model in Section 6 converges cleanly from every initialization with zero residual — by every solver diagnostic the run looks healthy — while representing no metastable state of the data and freezing its own training loss. Appendix G states the objection and our reply in full.*

EquiPhase forgoes exactly that certificate and replaces it with a weaker, physically meaningful one: dissipativity (Proposition 1) plus empirical convergence diagnostics reported per run. Section 6 shows both halves of this trade empirically: our verified-contraction baseline collapses precisely as Proposition 2 mandates, while EquiPhase recovers the multistable structure of real molecular data.

5 Experiment 1: Synthetic Double-Well with Analytical Ground Truth

We first validate on a 32-dimensional anisotropic double-well potential whose minima ($\pm\sqrt{\alpha}\, e_1$), saddle points, and barrier height (0.2275) are known in closed form, so every empirical claim can be scored against an analytical answer. The residual network v_{net} is a small MLP ($34 \to 64 \to 1$; 2305 parameters). Under a preregistered gate battery (seed 7777): the structural symmetry and identity gates hold to tolerances of order 10^{-9}; two attractors are recovered ($N = 2$) with contraction factors $\rho = 0.9626/0.9628$ against the corrected theoretical value 0.96273; and the measured escape barrier 0.230111 exceeds the analytical barrier 0.2275, consistent with the learned residual steepening the well walls. Two preregistered stress disclosures accompany these results: at the boundary setting $\alpha = 1.2$ one gate exceeds its per-α limit by 1.0×10^{-5}, and 1 of 100 novel-seed initializations diverges; both are reported as limits of the method rather than removed. Under the same protocol, an unconstrained vanilla DEQ fails to converge ($0/100$ initializations; residual $\approx 5.9 \times 10^{-3}$, $\rho \approx 0.9999$), and a monotone DEQ collapses to a single attractor with its training loss frozen — the same qualitative signature that reappears on molecular data in Section 6. Appendix B tabulates all values against their analytical targets.

6 Experiment 2: Alanine Dipeptide

Setup. We use the mdshare alanine dipeptide backbone-dihedral dataset (Computational Molecular Biology Group, FU Berlin (Noé group), 2017) (three independent 250 ns trajectories; 750,000 frames of (ϕ, ψ) in radians), pinned by SHA-256 and re-verified at every run. The potential V_θ is a small MLP over the periodic embedding $[\sin\phi, \cos\phi, \sin\psi, \cos\psi]$, trained by denoising score matching ($\sigma = 0.15$ rad, 3000 Adam (Kingma & Ba, 2015) steps, batch 4096) on five preregistered seeds. Attractors are found by deterministic gradient descent from a 24×24 grid; basins are assigned to macrostates (β, α_R, α_L) by preregistered angular regions.

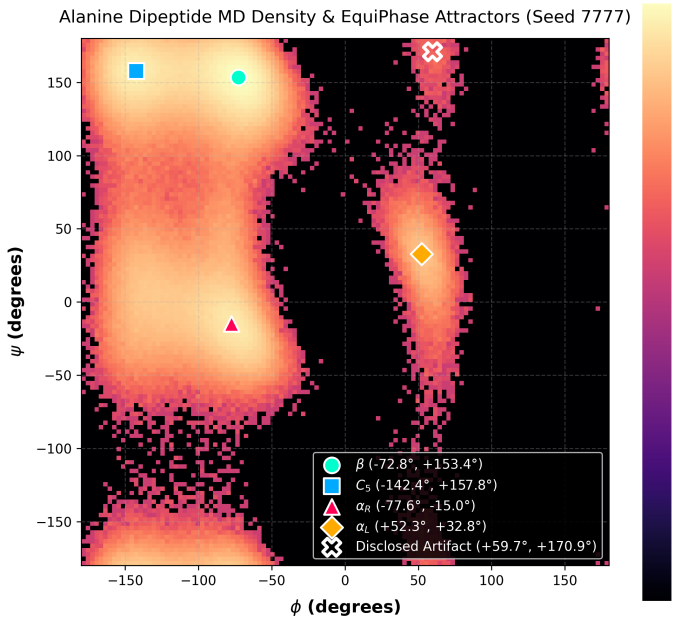


Figure 1: MD dihedral density (750k frames) with the five attractors recovered by EquiPhase (seed 7777) overlaid. Filled markers: physical states (β, C_5/β sub-mode, α_R, α_L); hollow marker: disclosed high-V artifact.

Preregistered gates. R1: attractors in both β and α_R; R2: locations within $\pm 25°$ (per axis, periodic) of data-derived anchors $(-72.5°, +152.5°)$ and $(-72.5°, -17.5°)$; R3: $V(\beta) < V(\alpha_R)$ with a $0.05\, k_B T$ margin; R4: baseline separation; R5 (exploratory): detection of the rare $\phi > 0$ state. Aggregation (fixed before execution): pass requires $\geq 4/5$ seeds.

Results. EquiPhase passes R1, R2, and R3 in 5/5 seeds (Figures 1 and 2; Table 1). Recovered basins reproduce the fine structure of the landscape: both β-branch sub-modes (near $(-73°, +153°)$ and $(-141°, +158°)$) and the α_R basin (near $(-78°, -15°)$) appear consistently across seeds. The learned free-energy gap $\Delta F(\alpha_R - \beta)$ ranges over 0.54–$0.84\, k_B T$ across seeds. Post-hoc observation (not a preregistered gate): this bracket contains the value $0.71\, k_B T$ obtained by an independent histogram estimate on the same data. In 5/5 seeds the model additionally places an attractor in the rare α_L region at approximately $(+53°, +31°)$ (R5), although this region holds only 3.0% of frames and was not isolated as a mode by our 72×72 density EDA; a second, high-V attractor near $(+58°, +175°)$ appears in all seeds and is disclosed as a probable shallow artifact (Section 7 states the identification criterion).

Baselines, reported in full. The vanilla score network (same budget, seed 7777) also places attractors near the physical basins, so the preregistered gate R4b fails, and we report this: in a two-dimensional collective-variable space, localization is easy. What the unconstrained field cannot provide is a potential: its mean absolute curl is 1.67, so no scalar V exists whose gradient matches it, and gate R3 is undefined for it. (A potential can of course be extracted from a non-conservative field post hoc, e.g., by Helmholtz projection; the resulting surface then depends on the projection and inherits no consistency guarantee. A sealed comparison against exactly such a projected baseline is preregistered as follow-up experiment E-A in Appendix D.) The monotone baseline, after correcting an auditor-side implementation error (documented in the audit ledger; the first version scaled its correction by 0.9π and was not a contraction), is a verified contraction (theoretical bound 0.95; empirical Lipschitz maximum 0.788) and behaves exactly as Banach's theorem requires: every one of 576 initializations converges (residual 0.0) to one non-physical fixed point (Figure 3), with training loss frozen at 0.712. Uniqueness guarantees and multistable physics are not jointly achievable in the autonomous regime; our results quantify what each side of that trade costs.

Table 1: Alanine dipeptide: preregistered gate outcomes and capability comparison. All numbers verbatim from sealed runs; "—" denotes undefined for that model class. R4b is a preregistered failure and is reported as such.

	EquiPhase DEQ	Vanilla score net	Monotone DEQ (verified)
R1 basins β, α_R	found 5/5 seeds	yes (seed 7777)	no (N=1 collapse)
R2 location $\pm 25°$	5/5 seeds	near basins (R4b fails)	no; $(-50.1°, +18.5°)$
R3 ordering $V(\beta)<V(\alpha_R)$	5/5 seeds	— (no potential)	—
$\Delta F(\alpha_R - \beta)$	0.54–0.84 k_BT	—	—
R5 rare α_L (3.0%)	5/5 seeds	—	no
Mean absolute curl of field	0 (by construction)	1.67	—
Training loss	0.019	—	0.712 (frozen)
Attractors (576 grid inits)	5 (4 physical + 1 disclosed)	—	1 (residual 0.0)
Lipschitz certificate	forgone by design	none	$0.788 < 0.95 < 1$

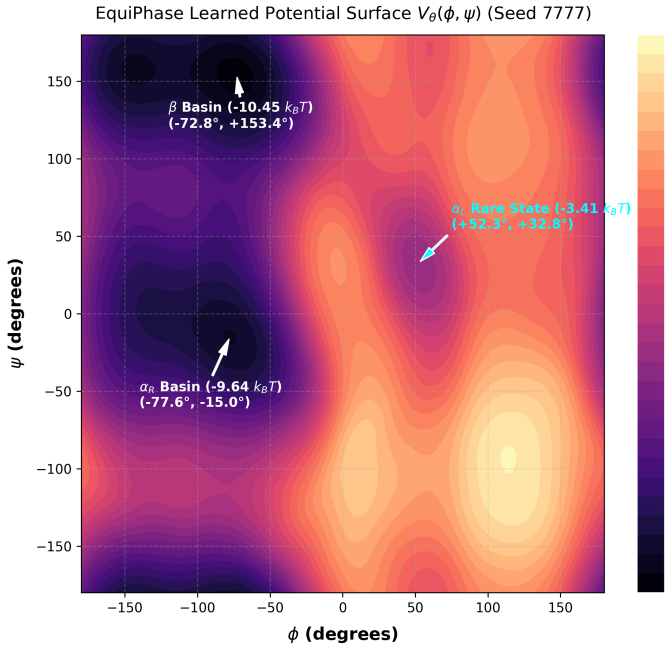


Figure 2: Learned potential $V_\theta(\phi, \psi)$, seed 7777. Basin minima and their depths, restated here for print legibility: β basin at $(-72.8°, +153.4°)$ with $V(\beta) = -10.45\, k_BT$; α_R basin at $(-77.6°, -15.0°)$ with $V(\alpha_R) = -9.64\, k_BT$; hence $\Delta F = +0.81\, k_BT$ with the correct ordering $V(\beta) < V(\alpha_R)$. The rare α_L state (3.0% occupancy) forms a distinct local minimum at $(+52.3°, +32.8°)$.

7 Discussion: Positioning, Value over Density Estimation, and Artifact Identification

Relation to the Markov-state-modeling toolchain. MSM/PyEMMA pipelines (Chodera & Noé, 2014; Scherer et al., 2015) and VAMPnets (Mardt et al., 2018; Wu & Noé, 2020) already identify metastable states from trajectories, so we state precisely what EquiPhase adds and what it does not. The MSM family answers a *kinetic* question: it discretizes or embeds the trajectory, estimates a transition operator, and extracts metastable sets and interconversion timescales from its spectrum. EquiPhase answers a *representational* one: it produces a single differentiable scalar field whose critical points *are* the model's states, on which relaxation from an arbitrary configuration, basin membership, and depth comparison are defined by construction rather than by a post-hoc clustering

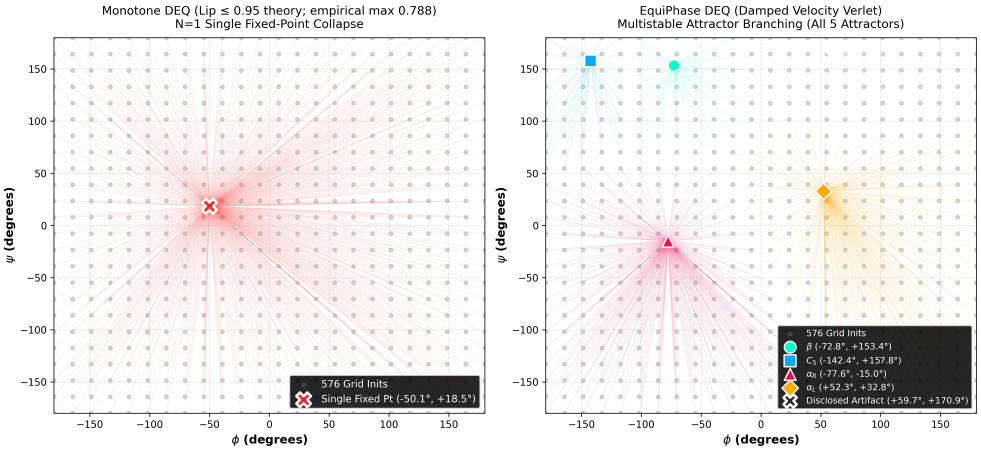


Figure 3: Verified Banach contraction (theoretical bound 0.95; empirical Lipschitz max 0.788): all 576 initializations collapse to one non-physical fixed point $(-50.1°, +18.5°)$ (left), versus EquiPhase multistable branching to physical basins (right; all five attractors shown).

step. The two are complementary, not competing: an MSM assigns no energy to an off-trajectory configuration and supports no gradient flow, while EquiPhase makes no kinetic claims whatsoever — transition rates and timescales are outside our gate battery by design (disclosure D11′). A sealed, preregistered comparison in which both toolchains are run on the identical hash-pinned dataset and scored on state recovery is specified as follow-up E-D′ in Appendix D; consistent with the FILL principle, no comparative numbers appear here because none have been produced under seal.

What the model buys over a histogram or KDE. In two dimensions with 750k frames, density estimation is not the hard part, and we do not claim otherwise; the post-hoc agreement of our ΔF bracket (0.54–$0.84\,k_BT$) with the independent histogram estimate ($0.71\,k_BT$) is reported as a consistency check, not as a win. The claims that a histogram cannot support are structural. First, *executability*: a binned density is not a dynamical system; EquiPhase is one, and its attractor set is provably identical to the critical-point set of V_θ (Appendix A), so "which basin does this configuration relax into" is answered by running the model, mesh-free and without a cell-assignment rule. Second, *consistency*: the field is curl-free by construction, so the depth comparison of gate R3 is well-defined — precisely the readout that the vanilla score baseline (curl 1.67) cannot support and that a histogram supports only at its chosen bin resolution. Third, and empirically the sharpest point in our results: *resolution where mass is scarce*. Our own preregistered 72×72 histogram EDA failed to isolate the α_L mode at 3.0% occupancy, while the learned potential places an attractor in that region in 5/5 seeds. We are careful about what this does and does not show: it does not show that no density estimator could find the state (a finer or adaptive estimator might), but it does show that the smooth parametric surface resolved structure that our preregistered mesh-based analysis of the same data missed — and the honest test of whether this advantage survives dimensionality is exactly experiment E-D of Appendix D.

How the disclosed artifact was identified, and the criterion's limit. Table 1 reports 5 attractors, disclosed as 4 physical plus 1 probable artifact, and a reviewer is entitled to ask by what rule the split was made — if the answer is "we looked at the data density," the independent value of attractor discovery is compromised. The rule actually applied is the preregistered one: basin assignment to macrostates is by the angular regions fixed in the preregistration (Appendix B), and the four attractors labeled physical are those falling in the β, C_5/β, α_R, and α_L regions, while the fifth, near $(+58°, +175°)$, falls in no preregistered macrostate region and sits at high V relative to the named basins — hence "probable shallow artifact," a model-internal designation that does not consult the raw density post hoc. We disclose the criterion's limit with equal clarity: region membership is a coarse classifier, and a preregistered depth-margin criterion (an attractor must be deeper than a margin fixed in advance, in units of k_BT, to be labeled a state) would be stronger. Committing to such a threshold now, after seeing the landscape, would be post-hoc; it is therefore specified as part

of the preregistration amendment for follow-up E-B in Appendix D, to be fixed before any new run executes (disclosure D10′).

8 Limitations and Disclosures

In accordance with our preregistered auditing protocol we disclose the scope boundaries, architectural assumptions, and methodological limits of this work. Items D0′–D7′ are the preregistered disclosures; D8′–D9′ were added during manuscript revision v2, and D10′–D11′ during revision v4. **The full, unabridged text of every item is reproduced verbatim in Appendix F**; nothing is withheld by the compression below.

- **D0′ (synthetic disclosures carried over):** the six preregistered disclosure items of the synthetic experiment apply to Section 5 in full.
- **D1′ (provided collective variables):** input is restricted to predefined 2D dihedrals (ϕ, ψ); this is not a test of unsupervised collective-variable discovery.
- **D2′ (no analytical prior on molecular data):** V_θ is learned alone for alanine dipeptide, so the prior layer of Section 3 is only partially applicable.
- **D3′ (1D barrier proxy scope):** the Phase-1 proxy measures intra-branch density fluctuation, not the 2D minimum-free-energy-path saddle; no quantitative barrier claim is made.
- **D4′ (single molecular system):** validation is exclusively on alanine dipeptide (750,000 frames); no generalization claim is made, and follow-up E-D is the preregistered path to lifting this boundary.
- **D5′ (full disclosure of exploratory α_L results):** α_L recovered at $(+52.3°, +32.8°)$ in 5/5 seeds, alongside a second high-V attractor at $(+59.7°, +170.9°)$ disclosed as a probable shallow artifact.
- **D6′ (auditor implementation Error #5):** the first monotone implementation scaled its update by $0.9\pi \approx 2.83$ and was not a contraction; corrected in Version 2 (bound 0.95; empirical $0.788 < 1$), which produced the verified $N = 1$ collapse.
- **D7′ (audit incident log):** all incidents INC-01 through INC-25 and the sealed script hash manifests are preserved in the anonymous repository.
- **D8′ (density–potential equivalence; v2):** V estimates the σ-smoothed negative log-density; R3 is a claim about the implied free-energy landscape, not about physical information beyond the data.
- **D9′ (baseline parity; v2):** both baselines share the main model's budget and seed; the monotone baseline's frozen loss 0.712 reflects the certified constraint, not an untuned implementation.
- **D10′ (artifact criterion coarseness; v4):** the physical/artifact split uses only macrostate region membership and relative V; a preregistered depth-margin criterion is deferred to the E-B amendment.
- **D11′ (no kinetic claims; v4):** no claims are made about transition rates, timescales, or pathways; the quantitative MSM comparison is confined to follow-up E-D′.

9 Conclusion

Convergence certificates built on global contraction are incompatible with multistable physics in the autonomous equilibrium regime: this is a theorem, and we made it an experiment. On real molecular data, an equilibrium model that trades the uniqueness certificate for a dissipative conservative structure recovers the metastable architecture of alanine dipeptide — including a state occupying 3.0% of the data — across all preregistered seeds, while a verified contraction collapses to a single non-physical point. We also report where structure does *not* help: an unconstrained score field locates the same basins, and what it cannot do is integrate to a potential. We believe the practice demonstrated here — preregistered gates (Nosek et al., 2018), hash-sealed evaluation, and full reporting of failed hypotheses and of our own auditing errors, in the spirit of the reproducibility literature (Pineau et al., 2021; Kapoor & Narayanan, 2023) — is as much a contribution as the architecture.

AI USE STATEMENT

This work used large language models (LLMs) extensively and under an explicit integrity protocol, which we disclose in full. **(i) Verification and evaluation code.** All scripts that produce quantitative claims (data audit, training-and-gate evaluation, and the corrected contraction re-test) were authored by an LLM acting as an external auditor, hash-sealed (SHA-256) before execution, and executed without modification; every reported number traces to the verbatim stdout of a sealed run. **(ii) Manuscript text.** Section drafts were generated by the auditor LLM from sealed outputs only; a second LLM agent performed repository execution, assembly, and documentation. **(iii) Bibliography.** Reference entries drafted by an LLM initially contained five defective entries (including one non-existent title); these were detected by independent verification against primary sources, corrected, and the incident recorded. All entries added in revision were re-verified against primary sources under the same protocol. **(iv) Oversight and accountability.** A human-maintained audit ledger records every detected LLM-produced falsehood, reconstruction, or protocol deviation in this project, including one auditor-side implementation error disclosed in Section 8. The human author made all research decisions, executed or directly supervised all runs, visually verified all figures, and takes full responsibility for the final content of this work, including all text, claims, and artifacts produced with AI assistance.

REPRODUCIBILITY STATEMENT

The dataset is the public mdshare alanine dipeptide dihedral archive (Computational Molecular Biology Group, FU Berlin (Noé group), 2017), pinned by SHA-256 (`F5AD3076...D4CD5717`) and re-verified inside every sealed script before any computation (hash mismatch aborts the run). Training and evaluation are fully deterministic given the preregistered seeds (7777, 1234, 2026, 31415, 65537); hyperparameters are frozen inside the sealed scripts, and all models are implemented in PyTorch (Paszke et al., 2019). An anonymous repository accompanying this submission contains: the preregistration (fixed before any training run, with one amendment sealed before first execution), all sealed scripts with their SHA-256 manifest, the verbatim stdout evidence logs of the two evaluation runs, and the audit incident ledger. The synthetic-system results in Section 5 are reproduced by the corresponding sealed audit script under seed 7777.

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A PROOFS

A.1 PROOF OF PROPOSITION 1

Write one step (1)–(3) as a map $(q_t, p_t) \mapsto (q_{t+1}, p_{t+1})$ and let $H = \nabla^2 V(q_{1/2})$ (symmetric) and $a = \Delta t^2/2m$. Differentiating, $\partial q_{1/2}/\partial q_t = I$ and $\partial q_{1/2}/\partial p_t = \frac{\Delta t}{2m} I$, so the block Jacobian $J = \left[\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right]$ has

$$A = I - aH, \quad B = \tfrac{\Delta t}{2m}\big[(2-\eta)I - aH\big], \quad C = -\Delta t\, H, \quad D = (1-\eta)I - aH.$$

With $\Omega = \left[\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right]$, the condition $J^\top \Omega J = c\,\Omega$ is equivalent to (i) $A^\top C$ symmetric, (ii) $B^\top D$ symmetric, (iii) $A^\top D - C^\top B = cI$. Since all four blocks are polynomials in the single symmetric matrix H, they commute and (i)–(ii) hold. For (iii),

$$\begin{aligned} A^\top D - C^\top B &= (I - aH)\big[(1-\eta)I - aH\big] + \Delta t\, H \cdot \tfrac{\Delta t}{2m}\big[(2-\eta)I - aH\big] \\ &= (1-\eta)I - aH - a(1-\eta)H + a^2H^2 + a(2-\eta)H - a^2H^2 \\ &= (1-\eta)I + aH\big[-1 - (1-\eta) + (2-\eta)\big] = (1-\eta)I, \end{aligned}$$

so $c = 1 - \eta$ and $J^\top \Omega J = (1-\eta)\Omega$ globally. Taking determinants, $\det J = (1-\eta)^d$ in $2d$-dimensional phase space, so volume contracts at the constant rate $(1-\eta)$ per step. □

A.2 PROOF OF PROPOSITION 2

This is the Banach fixed-point theorem: for a contraction g with constant $c < 1$ on a complete metric space, the iterates $z_{k+1} = g(z_k)$ form a Cauchy sequence for any z_0 (since $d(z_{k+1}, z_k) \le c^k d(z_1, z_0)$), their limit z^* is a fixed point by continuity, and uniqueness follows from $d(z^*, w^*) = d(g(z^*), g(w^*)) \le c\, d(z^*, w^*)$. □

A.3 FIXED POINTS COINCIDE WITH CRITICAL POINTS

At a fixed point of (1)–(3), $q_{t+1} = q_t$ and $p_{t+1} = p_t$. From (1) and (3), $q_{t+1} - q_t = \frac{\Delta t}{2m}(p_t + p_{t+1}) = \frac{\Delta t}{m} p_t$, forcing $p_t = 0$; then (2) with $p_{t+1} = p_t = 0$ and $\eta \in (0, 1)$ forces $\nabla_q V(q_t) = 0$. Conversely any $(q^\star, 0)$ with $\nabla_q V(q^\star) = 0$ is fixed. Hence the attractor set of the model is exactly the set of stable critical points of V. □

B GATE DEFINITIONS, FROZEN HYPERPARAMETERS, AND SYNTHETIC-SYSTEM TABLE

All values below restate configuration constants already reported in the main text or frozen inside the sealed scripts; no new measurements are introduced.

B.1 SYNTHETIC 32-D DOUBLE-WELL: ANALYTICAL TARGETS VS. MEASURED VALUES

Quantity	Analytical / theoretical	Measured	Status
Attractor count N	2	2	pass
Contraction factor ρ	0.96273 (corrected)	0.9626 / 0.9628	pass
Escape barrier	0.2275	0.230111	pass (steepened walls)
Symmetry / identity gates	tol. $\sim 10^{-9}$	held	pass
Boundary $\alpha = 1.2$ gate	per-α limit	excess 1.0×10^{-5}	disclosed
Novel-seed robustness	100/100	99/100 (1 diverged)	disclosed
Vanilla DEQ (uncontrolled)	converge	0/100; res. $\approx 5.9 \times 10^{-3}$, $\rho \approx 0.9999$	fail
Monotone DEQ	multistable	N=1 collapse; loss frozen	fail

Seed 7777; all numbers verbatim from the sealed audit run.

B.2 ALANINE DIPEPTIDE CONFIGURATION

Periodic embedding $[\sin\phi, \cos\phi, \sin\psi, \cos\psi]$; potential V_θ a small MLP; denoising score matching with $\sigma = 0.15\,\text{rad}$; 3000 Adam steps; batch 4096; preregistered seeds $\{7777, 1234, 2026, 31415, 65537\}$; attractor search by deterministic gradient descent from a 24×24 grid; basin assignment by preregistered angular macrostate regions; collapse test on a 576-point grid.

B.3 GATE BATTERY (VERBATIM FROM PREREGISTRATION)

R1: at least one attractor inside each of the β and α_R regions. R2: attractor location within $\pm 25^\circ$ per axis (periodic distance) of the data-derived anchors $(-72.5^\circ, +152.5^\circ)$ and $(-72.5^\circ, -17.5^\circ)$. R3: $V(\beta) < V(\alpha_R)$ with margin $0.05\, k_B T$. R4(a/b): separation from the contraction and vanilla-score baselines respectively; R4b failed and is reported. R5 (exploratory): detection of the rare $\phi > 0$ (α_L) state. Aggregation fixed before execution: pass requires $\geq 4/5$ seeds.

B.4 SYNTHETIC SYSTEM

32-dimensional anisotropic double-well; residual MLP $34 \to 64 \to 1$ (2305 parameters); analytical minima $\pm\sqrt{\alpha}\, e_1$; analytical barrier 0.2275; gate battery under seed 7777 with structural gates G1/G2′ at tolerance $\sim 10^{-9}$.

C BASELINE CONFIGURATIONS AND PARITY

Both baselines share the alanine-dipeptide training budget of the main model (3000 Adam steps, batch 4096, seed 7777) and are implemented in the same sealed evaluation scripts. The monotone baseline's contraction is established two ways: a theoretical bound of 0.95 by construction and an empirical Lipschitz maximum of 0.788 measured at run time; its first implementation, which scaled the update by $0.9\pi \approx 2.83$ and was not a contraction, is disclosed as audit Error #5 and superseded.

D PREREGISTERED FOLLOW-UP EXPERIMENTS (SPECIFICATIONS; SEALED RUNS PENDING)

This appendix specifies, at preregistration level of detail, the follow-up experiments whose sealed execution is pending at the time of this draft. Consistent with the FILL principle governing this project, the specifications below contain no experimental numbers beyond configuration constants fixed in advance; every result field is marked [TO-RUN] and will be filled verbatim from sealed stdout, or the corresponding experiment will be withdrawn from the paper before submission. Each experiment requires its own preregistration amendment, sealed (SHA-256) before first execution, extending the existing protocol.

E-A: Helmholtz-projected vanilla baseline (closes the post-hoc-extraction objection). *Motivation.* Section 6 argues that the vanilla score field (mean absolute curl 1.67) admits no consistent potential; the natural objection is that a potential can be extracted post hoc. E-A tests whether such an extracted surface supports the same readouts as the structural one. *Protocol* (fixed in the amendment before execution): from the trained vanilla score field of the existing sealed run (seed 7777, unchanged), compute the Helmholtz decomposition on the periodic (ϕ, ψ) domain via the spectral method, take the curl-free component, and integrate it to a scalar surface $\tilde{V}$; apply gates R1–R3 to $\tilde{V}$ with the identical anchors, margins, and macrostate regions of Appendix B; report the extracted $\Delta F(\alpha_R - \beta)$ and its position relative to the EquiPhase bracket; repeat under at least one alternative gauge/projection choice fixed in the amendment, to measure the projection-dependence asserted qualitatively in Section 6. *Result fields:* [TO-RUN] R1/R2/R3 verdicts for $\tilde{V}$; [TO-RUN] extracted ΔF per projection choice; [TO-RUN] inter-projection disagreement.

E-B: Damping and noise-scale sweeps, with a preregistered depth-margin criterion. *Protocol:* (i) η over a small grid fixed in the amendment; report attractor count, R1–R3 status, and divergence rate per η. (ii) $\sigma \in \{0.05, 0.10, 0.15, 0.25\}$ rad; report basin locations and the ΔF bracket per σ

(robustness of disclosure D8′). (iii) The amendment for E-B additionally fixes, before execution, the depth-margin threshold (in k_BT) referenced in disclosure D10′, and the physical/artifact classification of all recovered attractors is re-scored under it. *Result fields:* [TO-RUN] all sweep tables; [TO-RUN] re-scored attractor classification.

E-C: Baseline parity transcription. *Protocol:* transcribe verbatim from the existing sealed scripts and stdout (no new training) the parity fields enumerated in Appendix C: layer widths and parameter counts of both baselines, learning rate, solver iteration budget per initialization, stopping criterion, and residual threshold. This is a documentation run, not an experiment; it exists because the FILL principle forbids writing these values from memory. *Result fields:* [TO-RUN] parity table in Appendix C.

E-D: Higher-dimensional collective-variable system (lifts disclosure D4′). *Protocol:* a peptide system with more than two backbone dihedrals, selected and hash-pinned in the amendment; periodic embedding extended per-dihedral; the full gate battery re-derived for the new system with anchors computed by the same data-derived procedure as Appendix B and frozen before training; the same three-way comparison (EquiPhase / vanilla score / verified contraction) under identical budgets. E-D is the experiment in which mesh-based density estimation degrades by construction, and is therefore the decisive test of the value proposition argued in Section 7. *Result fields:* [TO-RUN] full gate table for the new system.

E-D′: Sealed comparison against the MSM/VAMPnet toolchain (scoped by disclosure D11′). *Protocol:* on the identical hash-pinned alanine dipeptide dataset, run a standard MSM pipeline (Scherer et al., 2015) and a VAMPnet (Mardt et al., 2018) with configurations fixed in the amendment; score both, and EquiPhase, on macrostate recovery against the preregistered regions; report agreement and disagreement without ranking, since the toolchains answer different questions (kinetic vs. representational) and the comparison is a positioning exhibit, not a contest. *Result fields:* [TO-RUN] state-recovery agreement table.

E Audit Ledger Summary

The anonymous repository preserves the full forensic record: the preregistration and its single pre-execution amendment; SHA-256 manifests of all sealed scripts; verbatim stdout logs of the two evaluation runs (from which every numeral in this paper is copied); and the incident ledger INC-01 through INC-25, including the bibliography incident (five defective LLM-drafted entries, detected and corrected) and auditor Error #5 (the invalid first contraction, superseded by the verified version reported in the main text). Amendments introduced for the follow-up experiments of Appendix D will be sealed and appended to the same manifest before any of those runs execute.

F Limitations and Disclosures — Unabridged Text

Section 8 compresses each item to two lines for length. This appendix reproduces the preregistered text in full. Items D0′–D7′ are the preregistered disclosures; D8′–D9′ were added during manuscript revision v2, and D10′–D11′ during revision v4; all are marked as such.

- **D0′ (Synthetic-System Disclosures Carried Over):** The six preregistered disclosure items of the synthetic experiment (conditional self-verification structure; IFT-to-unrolling deviation; integrator naming; the $\alpha = 1.2$ boundary excess; the $1/100$ divergent initialization; Set-A separation) apply to Section 5 in full.
- **D1′ (Provided Collective Variables):** Model input is restricted to predefined 2D backbone dihedrals (ϕ, ψ). This benchmark evaluates the model's capacity to represent multistable energy landscapes and relative free energies, not unsupervised collective variable discovery.
- **D2′ (Absence of Analytical Prior on Molecular Data):** While the synthetic system leverages a hybrid potential $V_{\text{total}} = V_{\text{base}} + v_{\text{net}}$, no analytical prior exists for alanine dipeptide; the learned potential is parameterized purely by the network, rendering the prior layer only partially applicable.

- **D3′ (EDA 1D Barrier Proxy Scope):** The 1D ϕ density barrier proxy derived in our Phase-1 EDA measures intra-branch density fluctuations rather than the 2D minimum free-energy path saddle point. Consequently, quantitative barrier claims are avoided, and only relative macrostate depth ordering (R3) is evaluated.
- **D4′ (Single Molecular System Boundary):** Empirical validation is performed exclusively on alanine dipeptide (750,000 frames). We make no claims regarding generalization to higher-dimensional protein dynamics or multi-protein datasets; follow-up E-D (Appendix D) is the preregistered path to lifting this boundary.
- **D5′ (Full Disclosure of Exploratory α_L Results):** All exploratory R5 results are reported in full. The model recovers the physical α_L state ($(+52.3°, +32.8°)$, 3.0% data population) across 5/5 seeds, alongside a second high-V attractor near $(+59.7°, +170.9°)$ disclosed as a probable shallow artifact.
- **D6′ (Auditor Implementation Error #5 Disclosure):** In Phase-2 baseline evaluation, the initial Monotone DEQ implementation scaled its update by $0.9\pi \approx 2.83$, violating the contractive theorem premise. This error was logged in the audit ledger (Error #5) and corrected in Version 2 (theoretical bound 0.95; empirical Lipschitz maximum $0.788 < 1$), which produced the verified $N = 1$ Banach collapse.
- **D7′ (Audit Incident Log):** All forensic audit incidents (INC-01 through INC-25) and sealed script hash manifests are preserved in the anonymous repository accompanying this submission.
- **D8′ (Density–Potential Equivalence; added in revision):** As stated in Section 3, the learned V estimates the σ-smoothed negative log-density over the provided collective variables. R3 is a claim about the free-energy landscape implied by the data, not about physical information beyond it; the contribution is the existence of a consistent, differentiable, curl-free surface supporting attractor dynamics, established under adversarial gates.
- **D9′ (Baseline Parity; added in revision):** The vanilla score baseline uses the same training budget and seed as reported; the monotone baseline's frozen loss (0.712) reflects the certified contraction constraint, not an untuned implementation — the certificate itself is what forbids the map from placing attractors at the data modes. Full baseline configurations are frozen inside the sealed scripts (Appendix C).
- **D10′ (Artifact Criterion Coarseness; added in revision v4):** The physical/artifact split of the five recovered attractors uses only preregistered macrostate region membership and relative V (Section 7); a stronger preregistered depth-margin criterion is deferred to the E-B amendment so that its threshold is fixed before, not after, any run that it will judge.
- **D11′ (No Kinetic Claims; added in revision v4):** EquiPhase makes no claims about transition rates, interconversion timescales, or pathways; the comparison to kinetic toolchains (Section 7) is representational only, and its quantitative counterpart is confined to the preregistered follow-up E-D′ with no numbers reported in this paper.

G THE CONTRACTION BASELINE: OBJECTION AND REPLY IN FULL

Remark 2 compresses the following for length.

One may object that no practitioner deliberately deploys a contractive DEQ as a multistable landscape model, so defeating one proves little. We disagree with the framing. Contraction and strong monotonicity are not exotic choices; they are the default well-posedness prescription of the DEQ literature (Winston & Kolter, 2020; Revay et al., 2020), the certificates a careful practitioner is instructed to install before trusting an equilibrium layer at all. Our claim is that this standard prescription, transplanted unmodified into the autonomous regime, fails silently: the collapsed model in Section 6 converges cleanly from every initialization with zero residual — by every solver diagnostic the run looks healthy — while representing no metastable state of the data and freezing its own training loss. The baseline therefore functions as a diagnostic exhibit of a failure mode that would not announce itself, quantified under the same sealed protocol as the proposed model, rather than as an adversary constructed to lose. The constructive half of the paper is precisely what one must give up (the global certificate) and what one can keep instead (dissipativity, Proposition 1) to make the autonomous regime well-behaved without forcing $N = 1$.